



Module 3

NFS Versions 4 and 4.1

About This Module



This module focuses on enabling you to do the following:

- Describe the features of NFS Version 4 (NFSv4)
- Describe the implementation of NFSv4.1 with parallel NFS (pNFS) on ONTAP software
- Describe the implementation of NFSv4 on NetApp storage
- Explain NFSv4 security using ONTAP software



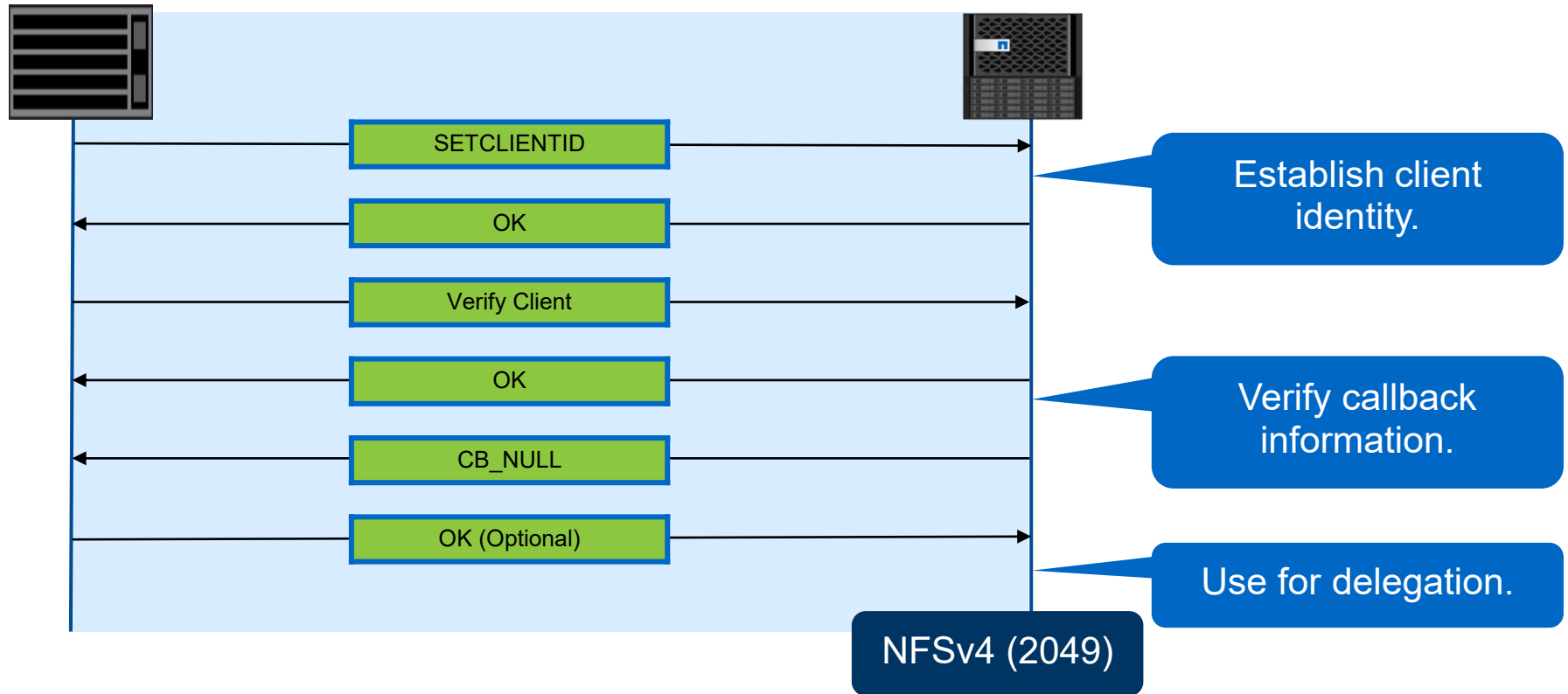
Lesson 1

Overview of NFSv4

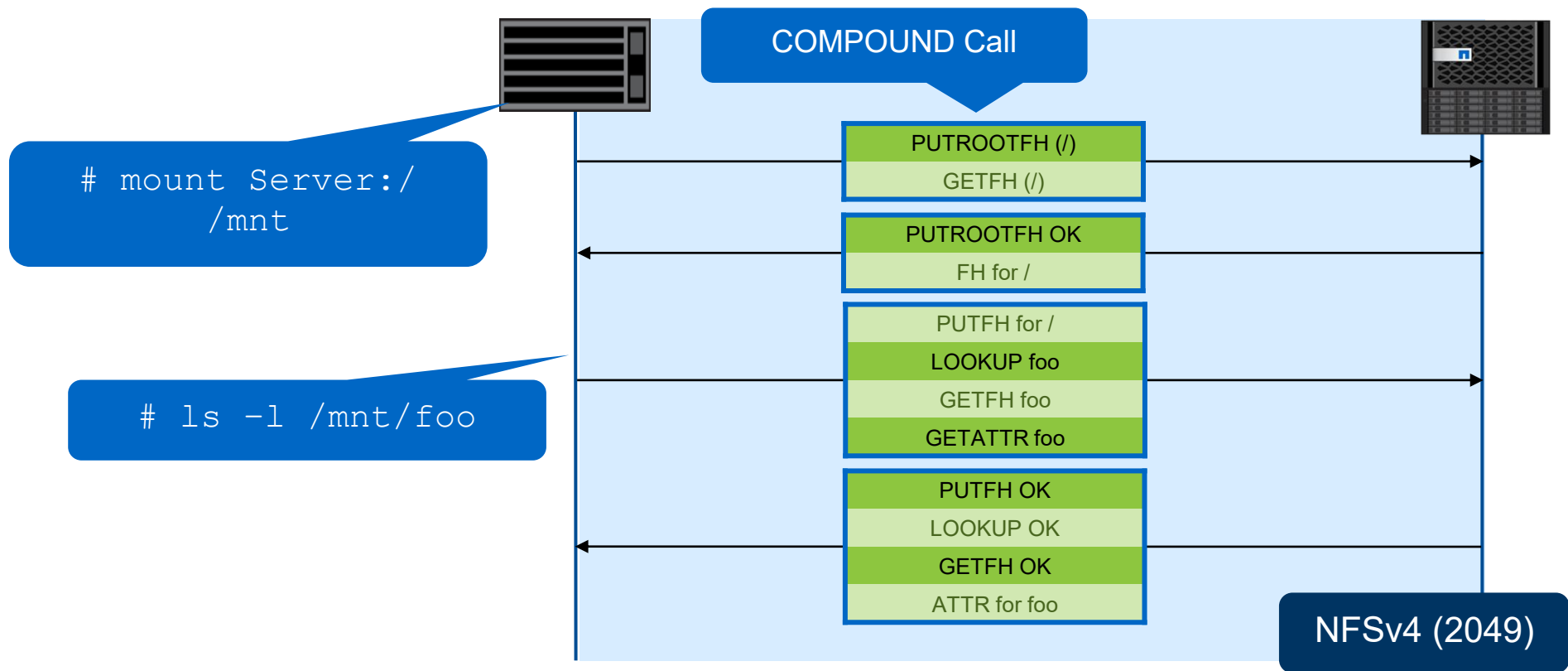
NFSv4 Features

- Requires TCP
- Can use all the authentication styles that NFS Version 3 (NFSv3) uses
- Is firewall-friendly:
 - Requires no mount, Network Lock Manager (NLM), Network Status Manager (NSM), or portmapper ports
 - Only one port (2049) is needed
- Supports UTF-8 for internationalization
- Has COMPOUND operations:
 - Packages several operations in one request
 - Reduces round-trip latency
- Pseudo file system

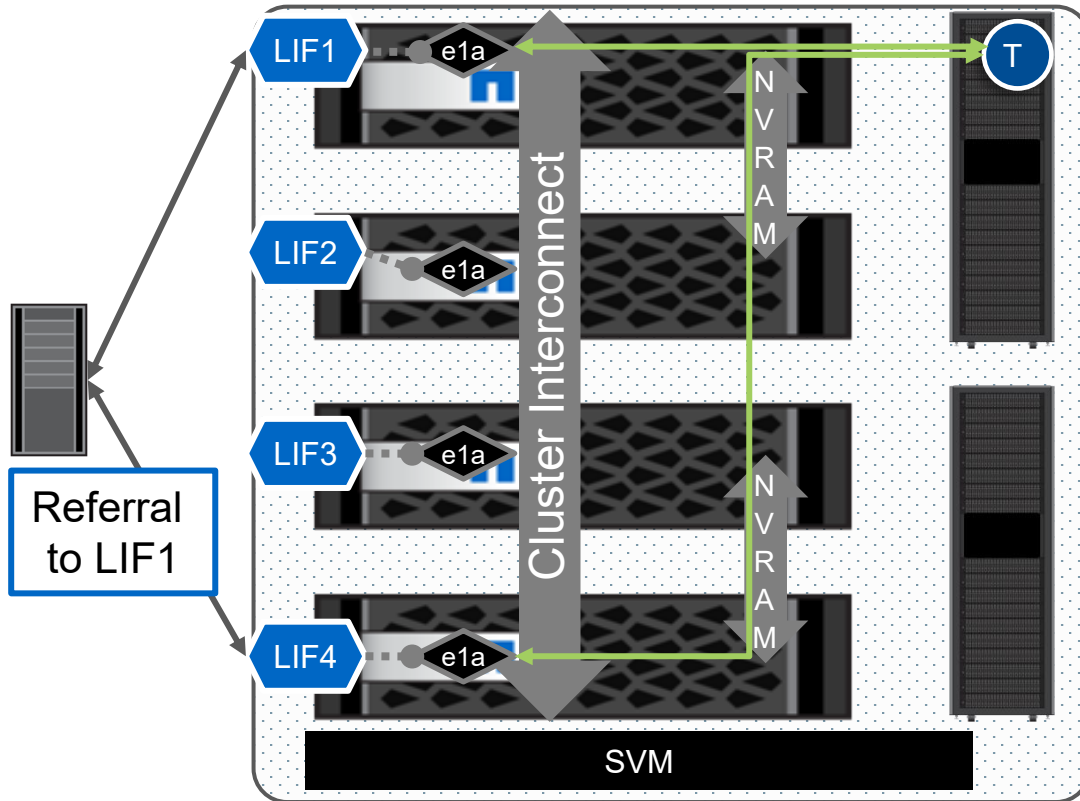
Connections



Mount Process



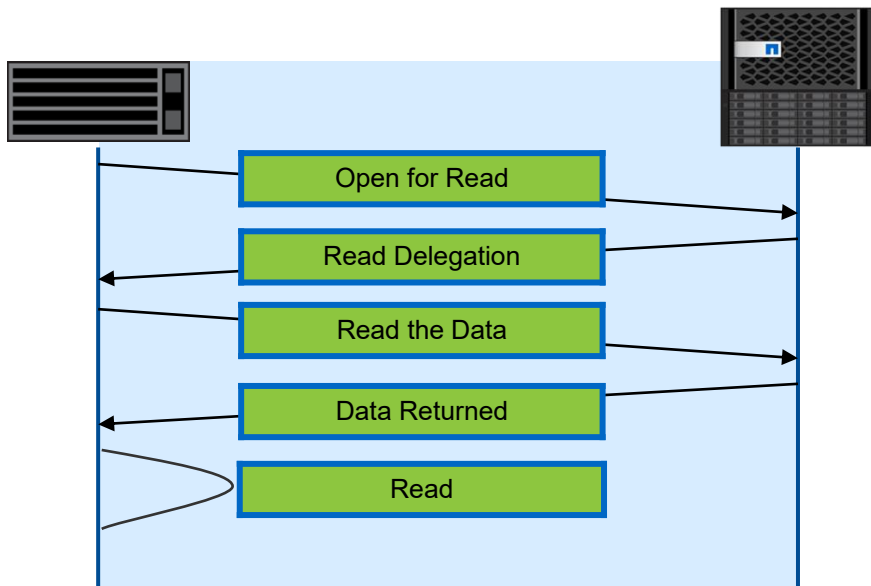
Referrals



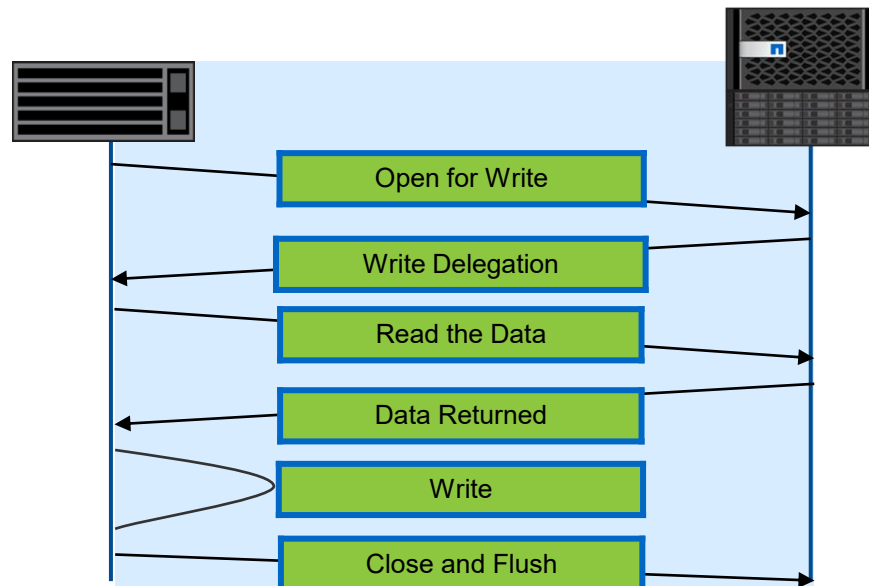
- Enables redirection to an optimized LIF during mount:
 - The volume T is stored on storage that is associated with the first node.
 - The initial request for T is through LIF4.
 - Access to T is over the cluster interconnect.
 - LIF4 issues a referral back to the client to use LIF1.
 - The client begins to use LIF1 to access T.
- Supports the following releases:
 - Clustered Data ONTAP 8.1 and later, including ONTAP 9 software
 - Red Hat Enterprise Linux 5.4 and later

Delegations

- Read delegation:
 - Prevents clients from writing or deleting the file
 - Can be recalled



- Write delegation:
 - Prevents other clients from accessing the file
 - Is not recalled by ONTAP software



Client-Server Lease Management

- Locks versus leases:
 - Versions earlier than NFSv4 require NSM and NLM to track locks.
 - NSM and NLM (and their ports) are not used in NFSv4.
 - NFSv4 issues leases, not locks.
- Timeout for locks:
 - Before NFSv4, locks were permanent.
 - NFSv4 requires clients to check in every 30 seconds, or the file lease is released.

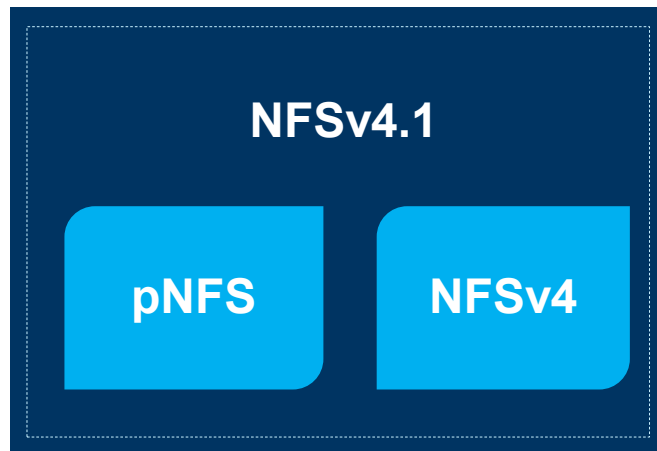


Use Cases

- NFSv4 has limited adoption because of client maturity:
 - Performance (not as fast as NFSv3)
 - Need for name-to-user ID (UID) mapping
- NFSv4 contains many improvements over earlier versions and should be evaluated for the following scenarios:
 - High-availability (HA) environments in which the new lease-management release locks within the cutover period
 - Interoperability between Windows-based and Linux-based hosts
 - Environments that require more structured access control list (ACL) security
 - Clustered client computers that involve many computational nodes that can use delegation
 - Metadata-intensive, read-only workloads

NFSv4.1

- Is a minor update to NFS Version 4 (NFSv4)
- Does not support ONTAP 9 directory delegations
- Supports the following features:
 - Reliable callbacks
 - Sessions
 - Layouts
 - pNFS
- Is supported in Linux:
 - Linux kernel 2.6.32 or later is required for basic NFSv4.1.
 - Red Hat Enterprise Linux 6.4 and later provide generally available support for NFSv4.1.
 - Other vendors support NFSv4.1 (contact the vendors for more details).

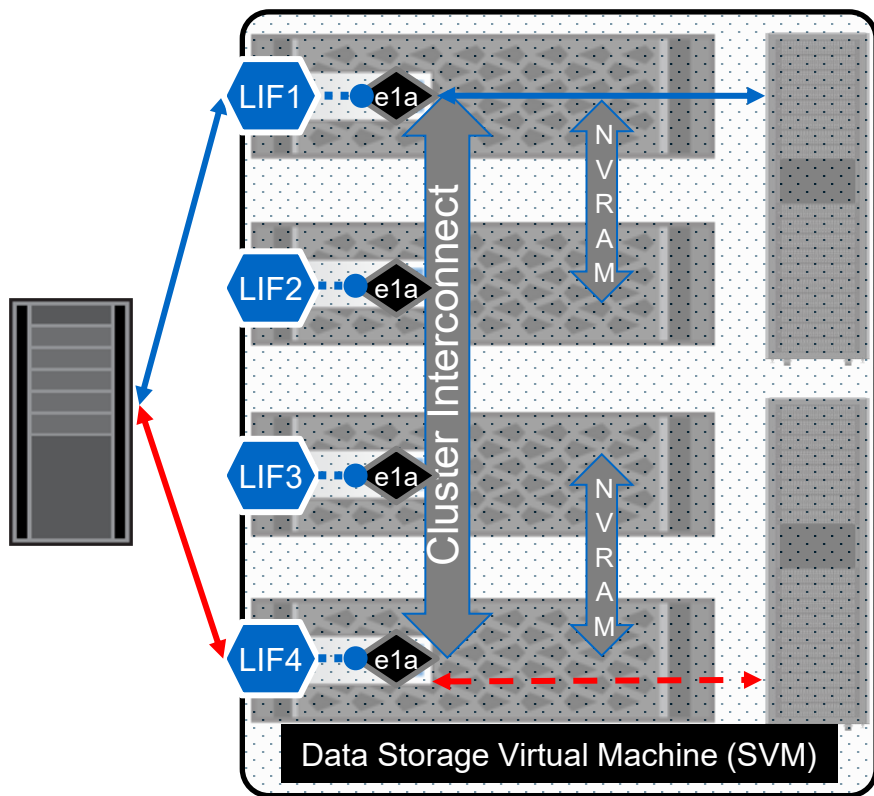




Lesson 2

NFSv4.1 with pNFS

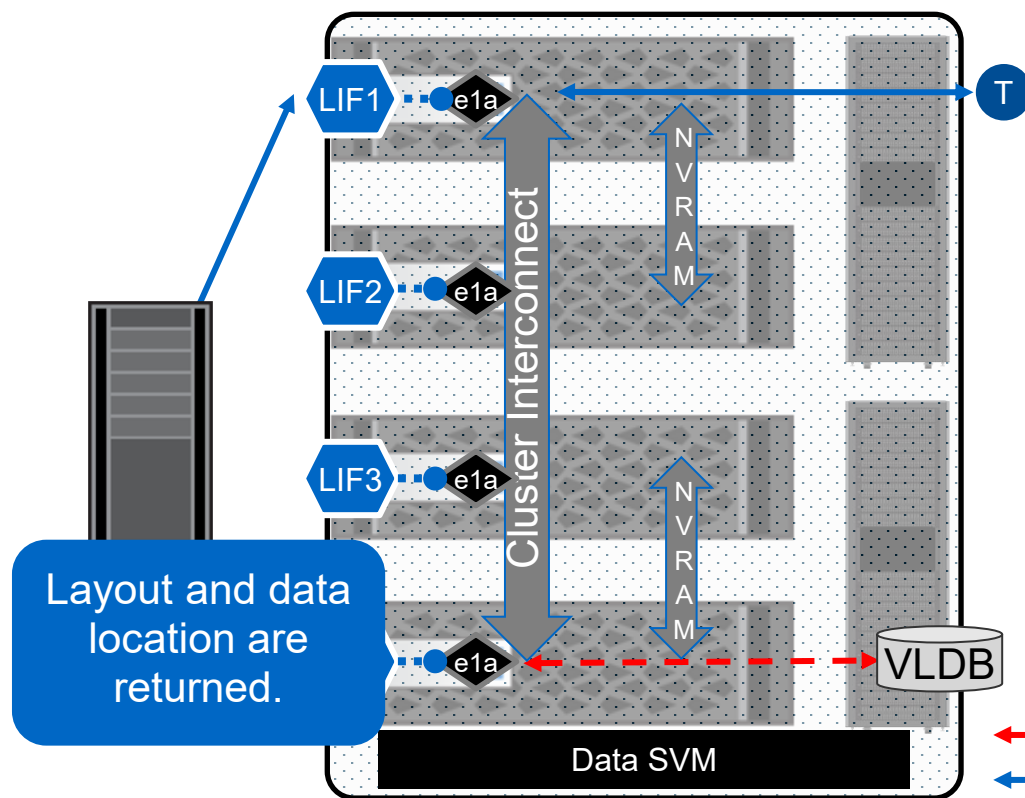
pNFS



- pNFS consists of the following servers:
 - Metadata
 - Data
- pNFS is supported in the following operating systems:
 - Clustered Data ONTAP 8.1 and later
 - Red Hat Enterprise Linux 6.4 and later
- In ONTAP 9 software, each node is a potential metadata server, because each node has a copy of the volume location database (VLDB).
- The client mount determines which node acts as a metadata server.
- Each data server should have a configured LIF.

↔ Metadata Server Paths
↔ Data Server Paths

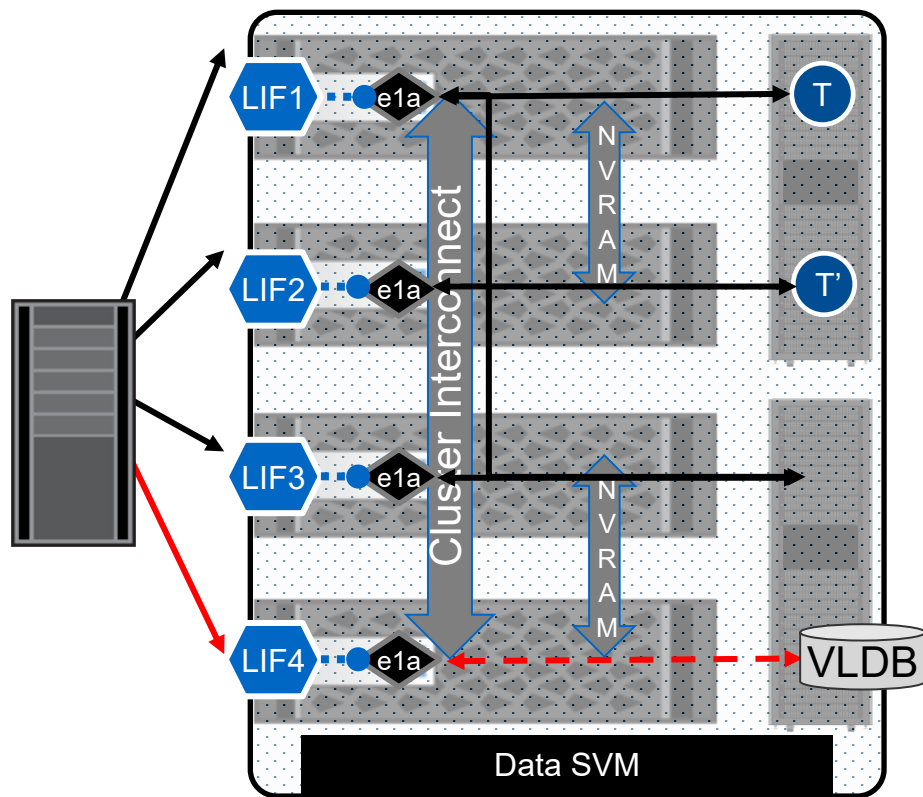
pNFS Access



- The client queries the metadata server for T volume layout and location.
- Access to T occurs over the cluster interconnect; the file is returned with location metadata.
- On subsequent reads, the client is redirected to use LIF1.
- The client begins to use LIF1 to access T.

↔ Metadata Server Paths
↔ Data Server Paths

pNFS Versus Referrals



- If the volume moves after the mount operation on the client, then the following events occur:
 - Referrals must remount so that access does not occur over the cluster interconnect.
 - pNFS updates the client access.
- Both data paths are optimized.
- Performance improves when data is spread across multiple physical servers.

↔ Metadata Server Paths
↔ Data Server Paths

Use Cases

- The main incentive for using NFSv4.1 is pNFS.
- Ideal uses include the following:
 - Efficient load balancing of business-critical applications, including many engineering applications
 - Effective resource use with different workload characteristics
 - Scaling and provision of storage to achieve quick SLAs for IT as a service (ITaaS) or platform as a service (PaaS)
 - Improved accessibility, reliability, and performance for hypervisors and databases



Not ideal

Common reasons to disable pNFS in an NFSv4.1 environment include:

- **Restricted Network Routing:** pNFS requires clients to directly access multiple, independent storage targets (Data LIFs). If firewall rules or network topologies prevent the client from reaching all of these IPs, read/write operations will hang.
- **Feature Incompatibility:** You cannot use pNFS and **NFSv4 referrals** simultaneously. If your environment uses referrals for mount redirection, pNFS must be turned off.
- **Complexity with Kerberos:** Using pNFS with Kerberos security requires Kerberos to be enabled on every single Data LIF, significantly increasing administrative overhead.
- **Performance with Newer Alternatives:** Instead of pNFS, many administrators now prefer using multipathing features like **nconnect** (NFSv4.1) or **session trunking** to split network traffic and overcome bottlenecks without the architectural complexity of pNFS.



Lesson 3

NFSv4 Implementation

NFSv4 Implementation

License, aggregate, and SVM tasks

1 Verify that the NFS license is added to the cluster.

2 Create an aggregate, if needed.

3 Create an SVM or use another SVM.



Enable NFS

ONTAP CLI

```
cluster1::*>vserver nfs create
    -vserver svmNFS
    -access true
    -v3 disabled
    -v4 enabled
    -udp false
    -tcp true
```

NFSv4 requires TCP.

```
cluster1::>vserver nfs show
```

Vserver	General Access	v3	v4	v4.1	UDP	TCP
-----	-----	-----	-----	-----	-----	-----
svmNFS	true	disabled	disabled	disabled	disabled	disabled

NFSv4 Configuration Options

ONTAP CLI

```
cluster1::*>vserver nfs modify
  -vserver svmNFS ...
  -v4.0-acl <enabled/disabled>
  -v4.0-read-delegation <enabled/disabled>
  -v4.0-write-delegation <enabled/disabled>
  -v4-fsid-change <enabled/disabled>
  -v4.0-referrals <enabled/disabled>
  -v4-validate-symlinkdata
  -v4-lease-seconds <integer>
  -v4-grace-seconds <integer>
  -v4-acl-preserve <enabled/disabled>
```

More Configuration Options

- LIF configuration
- Host, user, and user-mapping resolution configuration
- Export policies and rules configuration





Lesson 4

NFSv4 Security

ACLs

- ONTAP 9 support for NFSv4 ACLs (not POSIX ACLs)
- Benefits:
 - Improves NFS security and interoperability with SMB
 - Replaces the NFS limitation of 16 groups per user
- Permissions:
 - Are granted or denied on a per-user basis for each file object
 - Are inherited from parent to new files or subdirectories
- Access control entry (ACE) types:
 - ALLOW
 - DENY
- Maximum ACEs per ACL: 1024 (the default is 400)

```
cluster1::*>vserver nfs modify -v4-acl-max-aces #
```


ACE Permissions

Permission	Description
r	Read files or list directories
w	Write files or create files in directories
a	Append to files or create subdirectories
x	Execute files or change directories
d	Delete files or directories
D	Delete file or subdirectory
t	Read attributes
T	Write attributes
n	Read name attributes
N	Write name attributes
c	Read ACL
C	Write ACL
o	Change ownership
y	Enable clients to use synchronous I/O with server

NFSv4 Implementation

Client configuration

- Install `nfs4-acl-tools` (if required):

```
# yum install nfs4-acl-tools
```

- Mount the export:

```
# mount -t nfs4 -o acl system:/ /mnt/system
```

- Set ACL:

```
# nfs4_setfacl -a A::500:rxtncy foo
```

- Remove ACL:

```
# nfs4_setfacl --test -x 2 foo
## Test mode only - the resulting ACL for "/mnt/vsNFS-v4/foo":
A::student@example.com:rwxtnCy
A::OWNER@:rwatTnNcCy
D::OWNER@:x
A:g:GROUP@:rtncy
D:g:GROUP@:waxTC
A::EVERYONE@:rtncCy
D::EVERYONE@:waxT
# nfs4_setfacl -x 2 foo
```

- View ACL:

```
# nfs4_getfacl foo
```

Adds the `nfs4_setfacl` and `nfs4_getfacl` executables

Using UID and GID in NFSv4

GID means group ID

Use the following steps to implement:

1. Enable the use of numeric ID:

```
ONTAP CLI cluster1::*>vserver nfs modify... -v4-numeric-ids enabled
```

2. Both the client and server must agree on a pseudo domain name:

```
ONTAP CLI cluster1::>vserver nfs modify... -v4-id-domain <d>
```

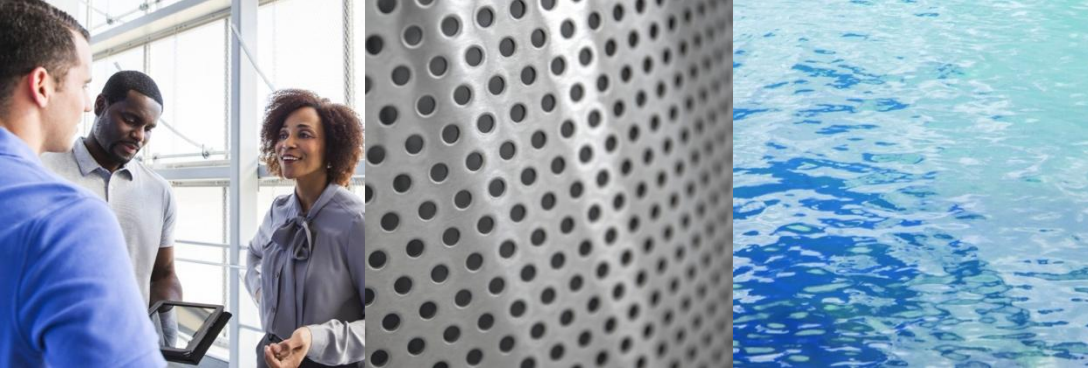
```
Linux Client # vi /etc/idmapd.conf# reboot
```

3. Create user and group accounts (case sensitive); either local users, Network Information Service (NIS), or (most often) Lightweight Directory Access Protocol (LDAP).

Snapshot Directories in NFSv4

- The Snapshot directory (.snapshot) is visible at every hierarchical level in NFSv3.
- In NFSv4, the .snapshot directory is not visible but is accessible from the client.

```
# mount -t nfs4 svm1:/ /mnt/svm1
# cd /mnt/svm1
# ls -la | grep snapshot
# cd .snapshot
# ls -la
drwxrwxrwx. 12 root root 4096 Apr 25 16:05 .
drwxrwxrwx. 3 root root 106496 Apr 24 16:01 ..
drwxrwxrwx. 4 root root 106496 Apr 18 14:50 daily.2013-04-24_0010
drwxrwxrwx. 2 root root 106496 Mar 12 19:54 weekly.2013-04-14_0015
drwxrwxrwx. 4 root root 106496 Apr 18 14:50 weekly.2013-04-21_0015
```



Take a Poll

Two-Question Quiz

Duration: 10-20 minutes

Poll Questions



1. Which two actions do COMPOUND operations enable in an NFSv4 environment? (Choose two.)
 - a. packaging of several operations in one request
 - b. scripting of operations
 - c. use of NLM and NSM to lock files
 - d. reduction of round-trip latency

Poll Questions: Answers



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- ✓ d. reduction of round-trip latency

Poll Questions



2. Which NFSv4 feature is not compatible with pNFS?
- a. delegation
 - b. referrals
 - c. lease
 - d. COMPOUND operations

Poll Questions: Answers



2. Which NFSv4 feature is not compatible with pNFS?

- a. delegation
- ✓ b. referrals
- c. lease
- d. COMPOUND operations

Poll Questions



1. Which protocol does NFSv4 require?
 - a. UDP
 - b. TCP
 - c. SCTP
 - d. FCP

Poll Questions: Answers



1. Which protocol does NFSv4 require?

- a. UDP
- ✓ b. TCP
- c. SCTP
- d. FCP

Poll Questions



1. Which two benefits are advantages of ACLs that use NFSv4 in an ONTAP 9 environment? (Choose two.)
 - a. increases AES 128 encryption to AES 256 encryption technologies
 - b. improves NFS security and SMB interoperability
 - c. replaces the limitation of the number of groups per user
 - d. all of the above

Poll Questions: Answers



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Poll Questions



2. What is the maximum number of ACE entries per NFSv4 ACL?
- a. 400
 - b. 1024
 - c. 512
 - d. 256

Poll Questions: Answers



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Module Summary



This module focused on enabling you to do the following:

- Describe the features of NFS Version 4 (NFSv4)
- Describe the implementation of NFSv4.1 with parallel NFS (pNFS) on ONTAP software
- Describe the implementation of NFSv4 on NetApp storage
- Explain NFSv4 security using ONTAP software



Complete an Exercise

Module 3: Configuring an SVM for NFSv4

Duration: 35 minutes

YOU ARE HERE

- ✓ NFS Fundamentals
- ✓ NFS Version 3
- ✓ NFS Versions 4 and 4.1
 - Multiprotocol Configuration
 - Monitoring and Managing NFS